INTERIM FEDERAL SPECIFICATION

CUTTER BITS, TOOL HOLDER

(CARBIDE TIP)

This Interim Federal specification was developed by the General Services Administration, Federal Supply Service, Standardization Division, Washington, D. C. 20406, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers cutter bits with brazed sintered carbide tips for general purpose use when installed in tool holders.

1.1.1 Federal specification coverage. Federal specifications do not include all varieties of the commodity as indicated by the title of the specification, or which are commercially available but are intended to cover only those generally used by the Federal Government.

1.2 Classification.

1.2.1. Styles. Cutter bits shall be of the following styles as specified (see 6.2).

Style AR 0° side cutting, right hand
Style AL 0° side cutting, left hand
Style BR 15° side cutting, right hand
Style BL 15° side cutting, left hand

1.2.2 Sizes. The size is expressed in terms of width, height, and length dimensions of the tool shank, and is represented by a dash number added to the style designation (see tables 1 and 2 and 6.2).

2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards, of the issues in effect on date of invitation for bids, or request for proposal form a part of this specification to the extent specified herein.

Federal Specifications:

QQ-S-633 - Steel Bars
PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-636 - Box, Fiberboard
PPP-B-665 - Boxes, Paperboard, metal stayed
PPP-B-676 - Boxes, Set-up, Paperboard

Federal Standards:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Agencies).
Fed. Std. No. 151 - Metals, Test Methods.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C., 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

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Military Specification:
MIL-P-116 - Preservation, Methods of.

Military Standards:
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. The tool body shall be steel and the cutting tip shall be sintered carbide.

3.2 Hardness. No specific hardness is required for the tool body. The hardness of the carbide tip shall be not less than Rockwell A 90.

3.3 Design. Style AR and AL cutter bits shall have a 0 degree side cutting angle, shall be similar to figure 1 and shall conform to the dimensions shown in table I. Style BR and BL cutter bits shall have a 15 degree side cutting angle, shall be similar to figure 2 and shall conform to the dimensions shown in table II. The cutting tip shall be brazed to the tool body and shall be capable of withstanding the test specified in 4.4.2.

Table I. Style AR and AL cutter bit dimensions

Style and Size	Tool Dimensions						Tip Dimensions			
	A Width ±0.000 -0.010	B Height ±0.000 -0.010	C Length ± 1/8	R Nose Rad.	F	H	Tip Over-Hang Max.	Thick. T	Width W	Length L
AR-4 AL-4	1/4	1/4	2	1/64 ±0.005	1/4 ±0.000 -0.015	1/4 ±0.000 -0.015	1/32	3/32	3/16	5/16
AR-5 AL-5	5/16	5/16	2-1/4	1/64	5/16	5/16	1/32	3/32	1/4	1/2
AR-6 AL-6	3/8	3/8	2-1/2	1/64	3/8	3/8	1/32	3/32	1/4	1/2
AR-7 AL-7	7/16	7/16	3	1/32	7/16	7/16	1/32	3/32	1/4	1/2
AR-8 AL-8	1/2	1/2	3-1/2	1/32	1/2	1/2	1/32	1/8	5/16	5/8
AR-10 AL-10	5/8	5/8	4	1/32 ±0.010	5/8 ±0.000 -0.020	5/8	1/32	5/32	3/8	3/4
AR-12 AL-12	3/4	3/4	4-1/2	1/32	3/4	3/4 ±0.000 -0.020	1/32	3/16	7/16	13/16
AR-16 AL-16	1	1	7	1/32	1	1	1/16	1/4	9/16	1

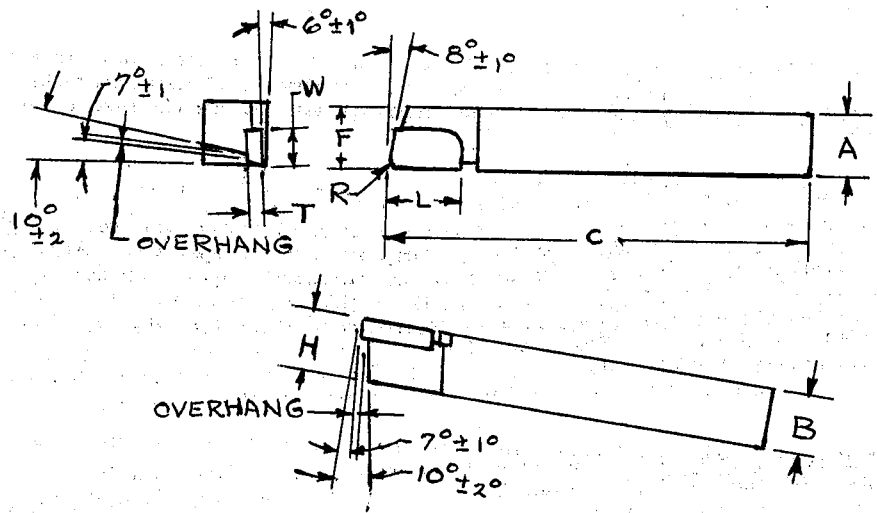


Figure 1. Style AR and AL cutter bits
(Style AR shown - style AL is opposite hand)

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Table II. Style BR and BL cutter dimensions

Style and Size	Tool Dimensions A Width	B Height	C Length	R Nose Rad.	F	H	Tip Over-Hang Max.	Tip Dimensions Thick. T	Width W	Length L
	±0.000 -0.010	±0.000 -0.010	± 1/8							
BR-4 BL-4	1/4	1/4	2	1/64	0.166	1/4	1/32	1/16	5/32	1/4
BR-5 BL-5	5/16	5/16	2-1/4	1/64 ±0.005	0.213	5/16	1/32	3/32	3/16	5/16
BR-6 BL-6	3/8	3/8	2-1/2	1/64	0.245	3/8 ±0.000 -0.015	1/32	3/32	1/4	1/2
BR-7 BL-7	7/16	7/16	3	1/32	0.290	7/16	1/32	3/32	1/4	1/2
BR BL-8	1/2	1/2	3-1/2	1/32	0.318	1/2	1/32	1/8	5/16	5/8
BR BL-10	5/8	5/8	4	1/32 ±0.010	0.398	5/8	1/32	5/32	3/8	3/4
BR-12 BL-12	3/4	3/4	4-1/2	1/32	0.507	3/4 ±0.000 -0.020	1/32	3/16	7/16	13/16
BR-16 BL-16	1	1	7	1/32	0.709	1	1/16	1/4	9/16	1

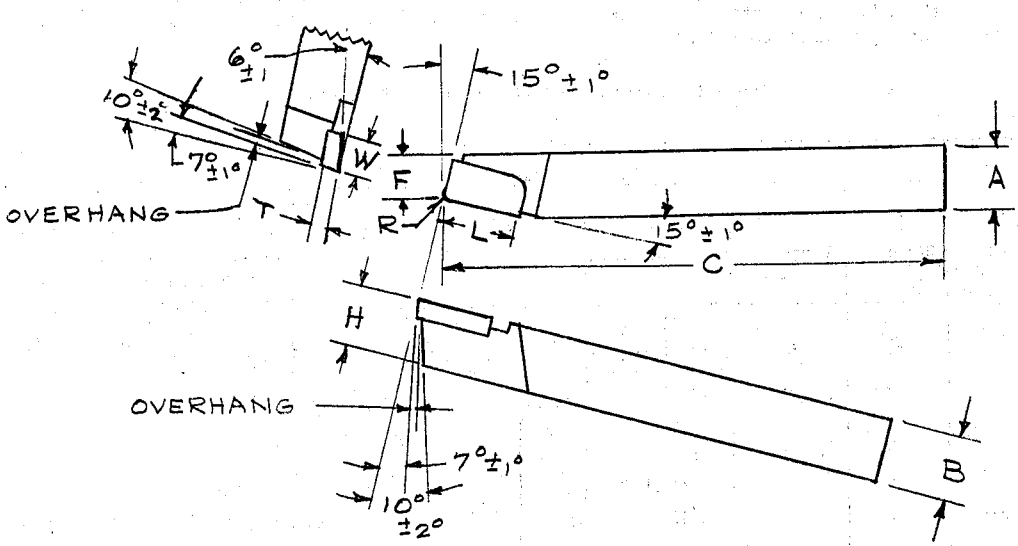


Figure 2. Style BR and BL cutter bits
(Style BR shown, Style BL is opposite hand)

3.4 Identification marking. Each cutter bit shall be marked on the tool body, in a plain and permanent manner with the name of the manufacturer, or with a trademark of known character so that the source of manufacture can be readily determined. Such markings may include the manufacturer's identification part number.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Sampling.

4.2.1 Inspection lot. All cutter bits offered for acceptance at one time shall be considered a lot for purposes of acceptance inspection.

4.2.2 Sampling for visual and dimensional inspection. A random sample shall be selected from each lot in accordance with MIL-STD-105 at inspection level II. The AQL shall be 1.5 percent defective.

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4.2.3 Sampling for acceptance testing. A random sample shall be selected from each lot in accordance with MIL-STD-105 at inspection level S-4. The AQL shall be 2.5 percent defective.

4.3 Inspection and tests.

4.3.1 Visual and dimensional inspection. Each sample selected in accordance with 4.2.2 shall be visually and dimensionally inspected to determine conformance to the requirements of this specification. Any bit containing one or more defect shall be classified a reject.

4.3.2 Inspection test. Each sample selected in accordance with 4.2.2 and determined to conform to 4.3.1 shall be tested as specified in 4.4.1 and 4.4.2. Any cutter bit that fails either test shall be considered a reject.

4.4 Test procedures.

4.4.1 Hardness test. Hardness test shall be conducted in accordance with method 243 of Federal Test Method Standard No. 151 to verify conformance to 3.2.

4.4.2 Performance test. This test shall be conducted on any metal turning lathe that has automatic feed controls. The test specimen shall be a section of round, cold rolled steel bar, 2 inches in diameter conforming to QQ-S-633, B1112 or C1119, annealed condition. The specimen shall be of sufficient length to permit the test cutter bit to complete cuts of 18 inches in traverse length. The cutter bit shall be secured in the tool holder in conformance with standard shop practices for carbide tools. The controls of the machine shall be set for a depth of cut of 0.250 inch plus or minus 0.005 inch and a surface speed of 200 F.P.M. Any cutter bit that fails to complete two cuts of 18 inch length shall be classified as a reject. Upon completion of the test the carbide tip of the cutter bit shall be inspected. The cutter bit shall be classified as a reject if the carbide tip contains spots due to wear, nicks, fractures or apparent looseness from tool shank.

4.5 Inspection of preparation for delivery requirements. An inspection shall be made to determine that the packaging, packing and marking comply with the requirements in section 5 of this specification. Defects shall be scored in accordance with table III. For examination of interior packaging the sample unit shall be one shipping container fully prepared for delivery, selected at random just prior to the closing operations. Sampling shall be in accordance with MIL-STD-105. Defects of closure listed shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 with an acceptable quality level (AQL) of 4.0 defects per hundred units.

Table III. Classification of preparation for delivery defects

Examine	Defects
Markings (exterior and interior)	Omitted; incorrect; illegible; improper size, location, sequence, or method of application.
Materials	Any component missing or damaged.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping, inadequate stapling. Distortion of container.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packing shall be level A, B or C as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning and drying. The bits shall be cleaned by process C-3 and dried by procedure D-4 in accordance with MIL-P-116.

5.1.1.2 Preservative application. The bits shall be coated with type P-8 preservative in accordance with MIL-P-116.

5.1.1.3 Unit packaging. Each bit shall be packaged method IC-1 in accordance with MIL-P-116.

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5.1.1.4 Intermediate packaging. Twelve bits of like description in 1/4 inch through 3/4 inch widths, and 2 inch to 4 1/2 inch lengths shall be intermediate packaged in quantities of 12. Six bits of like description in 1 inch widths and 6 inch lengths shall be intermediate packaged in a box conforming to PPP-B-566, PPP-B-665, or PPP-B-676.

5.1.2 Level B. The bits shall be preserved, and packaged as specified in 5.1.1.

5.1.3 Level C. The bits shall be packaged in accordance with the suppliers commercial practice.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. Intermediate packages of bits of like description in quantities as specified (see 6.2) shall be packed in a close-fitting box conforming to PPP-B-636, class weather-resistant. The box shall be closed, waterproofed, sealed, and strapped in accordance with the appendix to the box specification.

5.2.2 Level B. Intermediate packages of bits of like description in quantities as specified (see 6.2) shall be packed in a close-fitting box conforming to PPP-B-636, class domestic. The box shall be closed in accordance with the appendix to the box specification.

5.2.3 Level C. The bits shall be packed to insure carrier acceptance and safe delivery at destination in containers complying with the rules and requlations applicable to the mode of transportation.

5.3 Marking.

5.3.1 Civil agencies. In addition to markings required by the contract or order, the interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 Military activities. In addition to markings required by the contract or order, the interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES.

6.1 Intended use. The cutter bits described herein are intended for general purpose use in forming and shaping steel and steel alloy materials.

6.2 Ordering data. Purchasers should select the preferred options offered herein and include the following data in procurement documents:

(a) Title, number and date of this specification.

(b) Style and size of cutter bit. (See tables I and II)

(c) Level of preservation, packaging and packing required (see 5.1 and 5.2).

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